

2008 Ford Edge SE

2008 GENERAL INFORMATION Engine System - General Information - Edge & MKX

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SPECIFICATIONS

MATERIAL

Item	Specification	Fill Capacity
Dye-Lite® Gasoline Engine Oil Leak Detection Dye 164-R3700 (Rotunda)	-	29.6 ml (1 oz)
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A	-
Threadlock 262 TA-26	WSK-M2G351-A6	-

DESCRIPTION AND OPERATION

ENGINE

NOTE: When repairing engines, all parts must be contamination free. If contamination/foreign material is present when repairing an engine, premature engine failure may occur.

NOTE: Specifications show the expected minimum or maximum condition. Refer to the appropriate Engine article for the procedure.

NOTE: If a component fails to meet the specifications, it is necessary to refinish it or install a new component. Wear limits are provided as an aid to determine if the component can be refinished. A new component must be installed when any component fails to meet specifications and cannot be refinished.

NOTE: This service information contains information, steps and procedures that may not be specific to this engine.

This service information covers general procedures and diagnosis and testing of the engine system, except for exhaust emission control devices, which are covered in the **INTRODUCTION - GASOLINE ENGINES** article.

The engine incorporates the following features: Refer to the appropriate Engine article for the procedure.

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- Crankcase ventilation or breather system
- Exhaust emission control system
- Evaporative Emission (EVAP) control system

Some engines incorporate a fail-safe cooling system. Refer to the appropriate Engine article for the procedure.

The engine, fuel system, ignition system, emissions system and exhaust system all affect exhaust emission levels and must be maintained according to the maintenance schedule. Refer to the scheduled Maintenance Guide.

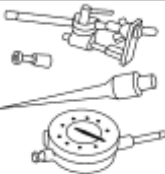
Correct engine identification is required to order parts. Refer to the appropriate Engine article for the procedure.

For complete vehicle and engine identification codes, refer to **IDENTIFICATION CODES** article.

DIAGNOSTIC TESTS

ENGINE

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1300-A	12 Volt Master UV Diagnostic Inspection Kit	164-R0756 or equivalent (Leak Detector)
 ST1214-A	Dial Indicator Gauge with Holding Fixture	100-002 (TOOL-4201-C) or equivalent
 ST1298-A	Engine Cylinder Leak Detection/Air Pressurization Kit	014-00708 or equivalent
 ST2312-A	EngineEAR	107-R2100 or equivalent

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 ST2311-A	EngineEAR/ChassisEAR	107-R2102 or equivalent
 ST1296-A	Oil Pressure Gauge	303-088 (T73L-6600-A)
 ST1299-A	Quick Disconnect Compression Tester	134-R0212 or equivalent
 ST1297-A	Vacuum/Pressure Tester	164-R0253 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS)	software with appropriate hardware, or equivalent scan tool

MATERIAL

Item	Specification
Dye-Lite® Gasoline Engine Oil Leak Detection Dye 164-R3700 (Rotunda)	-
Motorcraft SAE 5W-20 Premium Synthetic Blend Motor Oil XO-5W20-QSP (US); Motorcraft SAE 5W-20 Super Premium Motor Oil CXO-5W20-LSP12 (Canada); or equivalent	WSS-M2C930-A

There are 2 diagnostic paths that can be followed depending on the type of engine concern. Carry out Inspection and Verification - Engine Performance or Inspection and Verification - NVH.

Inspection and Verification - Engine Performance

1. Verify the customer concern by operating the engine to duplicate the condition.

2. Visually inspect for obvious signs of mechanical damage. Refer to the following chart.

VISUAL INSPECTION CHART

Mechanical
• Engine coolant leaks • Engine oil leaks • Fuel leaks • Damaged or severely worn parts • Loose mounting bolts, studs and nuts

3. If the inspection reveals obvious concerns that can be readily identified, repair as necessary.

NOTE: **Make sure to use the latest scan tool software release.**

4. If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC).

NOTE: **The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.**

5. If the scan tool does not communicate with the VCM:

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM.
- refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition key is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
- If the network test passes, retrieve and record continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the PCM.

9. If the DTCs retrieved are related to the concern, go to the DTC Chart, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.

10. If no DTCs related to the concern are retrieved, go to **SYMPTOM CHART - ENGINE** table.

Inspection and Verification - NVH

1. NVH symptoms should be identified using the diagnostic tools and techniques that are available. For a

list of these techniques, tools, an explanation of their uses and a glossary of common terms, refer to **NOISE, VIBRATION AND HARSHNESS** article.

2. Verify the customer concern by operating the engine to duplicate the condition.
3. Check the engine oil level and check the oil for contamination. Low engine oil level or contaminated oil are a common cause of engine noise. If the oil is contaminated, the source of the contamination must be identified and repaired as necessary.
4. Visually inspect for obvious signs of mechanical damage. Refer to the following chart.

VISUAL INSPECTION CHART

Mechanical
• Loose mounting bolts, studs and nuts • Damaged or leaking powertrain mounts • Damaged or disconnected vacuum hoses • Obstruction of cooling fan • Obstruction of Front End Accessory Drive (FEAD) • Damaged or disconnected air intake components

5. If the inspection reveals obvious concerns that can be readily identified, repair as necessary.

NOTE: **Make sure to use the latest scan tool software release.**

6. If the cause is not visually evident, connect the scan tool to the DLC.

NOTE: **The VCM LED prove out confirms power and ground from the DLC are provided to the VCM.**

7. If the scan tool does not communicate with the VCM:
 - check the VCM connection to the vehicle.
 - check the scan tool connection to the VCM.
 - refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no power to the scan tool.
8. If the scan tool does not communicate with the vehicle:
 - verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
9. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record continuous memory DTCs.
10. Clear the continuous DTCs and carry out the self-test diagnostics for the PCM.

11. If the DTCs retrieved are related to the concern, go to the DTC Chart, refer to **MULTIFUNCTION ELECTRONIC MODULES** article.
12. If no DTCs related to the concern are retrieved, continue the inspection and verification if a noise concern is related to the engine. For vibration concerns and noise concerns such as powertrain mounts, air intake system and starter Go to **SYMPTOM CHART - NVH** table.

In some cases, a noise may be a normal characteristic of that engine type. In other cases the noise may require further investigation. Comparing the noise to a similar year/model vehicle equipped with the same engine will aid in determining if the noise is normal or abnormal.

Once a customer concern has been identified as an abnormal engine noise, it is critical to determine the location of the specific noise. Use the EngineEAR/ChassisEAR or stethoscope (the noise will always be louder closer to the noise source) to isolate the location of the noise to one of the following:

- Fuel injector(s)
- Upper end of engine
- Lower end of engine
- Front of engine
- Rear of engine

Fuel injector noise - A common source of an engine ticking noise can be related to the fuel injector(s). This is normal engine noise that can be verified by listening to another vehicle. If the injector noise is excessive or irregular, use the EngineEAR/ChassisEAR or stethoscope to isolate the noise to a specific fuel injector.

Upper end engine noise - A common source of upper end engine noise (ticking, knocking or rattle) include the camshaft(s) and valve train. Upper end engine noise can be determined using the EngineEAR/ChassisEAR or stethoscope on the valve cover bolts. If the noise is loudest from the valve cover bolts, then the noise is upper end. The EngineEAR/ChassisEAR or stethoscope can be used to further isolate the noise to the specific cylinder bank and cylinder. Removal of the valve covers will be required to pinpoint the source of the noise.

Lower end engine noise - A common source of lower end engine noise (ticking or knocking) include the crankshaft, connecting rod(s) and bearings. Lower end noises can be determined by using the oil pan or cylinder block lug bosses. If the noise is loudest from these areas, then the noise is lower end. If an engine noise is isolated to the lower end, some disassembly of the engine may be required to inspect for damage or wear.

Front of engine noise - A common source of noise from the front of the engine (squeal, chirp, whine or hoot) is the Front End Accessory Drive (FEAD) components. To isolate FEAD noise, carry out the Engine Accessory Test, refer to **NOISE, VIBRATION AND HARSHNESS** article.

Some other noises from the front of the engine (ticking, tapping or rattle) may be internal to the engine. Use the EngineEAR/ChassisEAR or stethoscope on the engine front cover to determine if the noise is internal to the engine. Removal of the engine front cover may be necessary to inspect internal engine components.

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Rear of engine noise - A common source of noise from the rear of the engine (knocking) is the flywheel/flexplate. Inspection of the flywheel/flexplate will be necessary.

If equipped, the Rear End Accessory Drive (READ) can also be a source of noise from the rear of the engine (squeal or chirp). The READ consists of the coolant pump and belt.

Some engines have timing drive components at the rear of the engine and may be the source of noise (ticking, knocking or rattle). Use the EngineEAR/ChassisEAR or stethoscope on the rear of the engine if the noise is suspected to be internal to the engine. Some disassembly of the engine may be required to inspect for damage or wear.

13. After the noise is localized, note the characteristics of the noise, including type of noise, frequency and conditions when the noise occurs and Go to **SYMPTOM CHART - NVH** table.

Symptom Chart - Engine Performance

SYMPTOM CHART - ENGINE PERFORMANCE

Condition	Possible Sources	Action
• Difficult starting	• Inoperative or damaged ignition system • Air or vacuum leak • Inoperative or damaged fuel system • Inoperative or damaged starting system • Damaged charging system/battery • Burnt valve • Worn piston • Worn piston rings • Worn cylinder • Damaged head gasket • Inoperative or damaged cooling system • Fail-safe cooling invoked (if equipped)	• Refer to the appropriate Engine article for the procedure. REFER to the <u>Introduction - Gasoline Engines</u> article. • REFER to <u>CHARGING SYSTEM - GENERAL INFORMATION</u> article. • INSTALL a new valve in the cylinder head. TEST the system for normal operation after the repair. • INSTALL a new short block. TEST the system for normal operation after the repair. • INSTALL a new cylinder head gasket. TEST the system for normal operation after the repair. • REFER to <u>ENGINE COOLING</u> article. • REFER to the <u>Introduction - Gasoline Engines</u> article.

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• Poor idling	• Vacuum leaks • Inoperative or damaged ignition system • Inoperative or damaged cooling system • Inoperative or damaged fuel system • Fail-safe cooling invoked (if equipped) • Incorrect valve clearance • Incorrect valve-to-valve seat contact • Damaged head gasket	• Refer to the appropriate Engine article for the procedure. REFER to the <u>Introduction - Gasoline Engines</u> article. • REFER to <u>ENGINE IGNITION - 3.5L</u> article. • Refer to the appropriate Engine article for the procedure. • REFER to the <u>Introduction - Gasoline Engines</u> article. • ADJUST valve clearance. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair. • INSTALL a new cylinder head. TEST the system for normal operation after the repair. • INSTALL a new cylinder head gasket. TEST the system for normal operation after the repair.
• Engine runs rough	• Inoperative or damaged fuel system • Air or vacuum leaks • Inoperative or damaged cooling system • Inoperative or damaged ignition system • Fail-safe cooling invoked (if equipped) • Burnt or sticking valve • Weak or broken valve spring • Carbon accumulation in combustion chamber	• Refer to the appropriate Engine article for the procedure. • REFER to the <u>Introduction - Gasoline Engines</u> article. • INSTALL a new valve in the cylinder head. TEST the system for normal operation after the repair. • INSTALL a new valve spring. TEST the system for normal operation after the repair. • ELIMINATE carbon buildup. TEST the system for normal operation after the repair.
• Excessive oil	• Leaking oil	• REPAIR oil leakage. TEST the system for

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consumption	• Inoperative PCV system • Incorrect oil • Worn valve stem seal • Worn valve stem or valve guide • Sticking piston rings • Worn piston ring groove • Worn piston or cylinder	normal operation after the repair. • REPAIR or INSTALL new components as necessary. TEST the system for normal operation after the repair. • CHANGE oil to correct specification. • INSTALL a new valve stem seal. TEST the system for normal operation after the repair. • INSTALL a new cylinder head. TEST the system for normal operation after the repair. • INSTALL a new short block. TEST the system for normal operation after the repair.
• Oil in coolant	• Leaking head gasket • Leaking oil cooler (if equipped) • Damaged cylinder block • Damaged cylinder head	• INSPECT the engine components. INSTALL new engine components as necessary. Refer to the appropriate Engine article for the procedure. TEST the system for normal operation after the repair.
• Coolant in oil	• Leaking head gasket • Leaking oil cooler (if equipped) • Damaged cylinder block • Damaged cylinder head • Damaged coolant pump/seal/gasket	• INSPECT the engine components. INSTALL new engine components as necessary. Refer to the appropriate Engine article for the procedure. TEST the system for normal operation after the repair.
• Insufficient power	• Inoperative or damaged ignition system • Air intake system blockage • Lubrication system blockage • Inoperative or damaged fuel system • Oil level too high	• Refer to the appropriate Engine article for the procedure. REFER to the <u>Introduction - Gasoline Engines</u> article. • DRAIN oil to correct level. TEST the system

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- Incorrect engine oil
- Excessive accessory drive belt loading
- Inoperative or damaged cooling system
- Fail-safe cooling invoked (if equipped)
- Damaged or plugged exhaust system
- Incorrect tire size
- Dragging brakes
- Slipping transmission
- Incorrect valve clearance
- Worn or damaged valve tappet
- Damaged valve guide
- Compression leakage at valve seat
- Seized valve stem
- Weak or broken valve spring
- Worn or damaged cam
- Damaged head gasket
- Cracked or distorted cylinder head
- Damaged, worn or sticking piston ring(s)

- for normal operation after the repair.
- INSTALL correct specification engine oil. TEST the system for normal operation after the repair.
- Refer to the appropriate Engine article for the procedure.
- REFER to the **Introduction - Gasoline Engines** article.
- INSPECT exhaust system.
- REFER to **SUSPENSION SYSTEM - GENERAL INFORMATION** article.
- REFER to **BRAKE SYSTEM - GENERAL INFORMATION** article.
- REFER to **AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50** article.
- ADJUST valve clearance. REFER to **ENGINE - 3.5L** article. TEST the system for normal operation after the repair.
- INSTALL a new valve tappet. TEST the system for normal operation after the repair.
- INSTALL a new cylinder head assembly. TEST the system for normal operation after the repair.
- INSTALL a new valve spring. TEST the system for normal operation after the repair.
- INSTALL a new camshaft. TEST the system for normal operation after the repair.
- INSTALL a new head gasket. TEST the system for normal operation after the repair.
- INSTALL a new cylinder head assembly. TEST the system for normal operation after the repair.
- INSTALL a new short block. TEST the

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- Worn or damaged piston

system for normal operation after the repair.

Symptom Chart - NVH

NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to **NOISE, VIBRATION AND HARSHNESS** article. Since it is possible that any one of multiple systems may be the cause of the symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to **NOISE, VIBRATION AND HARSHNESS** article for the next likely system and continue diagnosis.

SYMPTOM CHART - NVH

Condition	Possible Sources	Action
• Drone type noise	• Powertrain mount(s)	• CARRY OUT the <u>Powertrain/Drivetrain Mount Neutralizing</u> procedure. TEST the system for normal operation after the repair.
• Drumming noise - occurs inside the vehicle during idle or high idle, hot or cold. Very low-frequency drumming is very RPM dependent	• Engine vibration excites the body resonances inducing interior noise	• CARRY OUT the <u>Powertrain/Drivetrain Mount Neutralizing</u> procedure. TEST the system for normal operation after the repair.
• Engine drumming noise - accompanied by vibration	• Powertrain mount(s)	• CARRY OUT the <u>Powertrain/Drivetrain Mount Neutralizing</u> procedure. TEST the system for normal operation after the repair.
• Rattle - occurs at idle or at light acceleration from a stop	• Powertrain mount(s)	• CHECK the powertrain mounts for damage. INSTALL new mounts as necessary. For engine, REFER to <u>ENGINE - 3.5L</u> article. For transaxle, REFER to <u>AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50</u> article. TEST the system for normal operation after the repair.
• Whine/moan type noise - pitch increases or	• Powertrain mount(s)	• CHECK the powertrain mounts for damage. INSTALL new mounts as necessary. For engine, REFER to <u>ENGINE - 3.5L</u> article. For transaxle, REFER to

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changes with vehicle speed		<u>AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50</u> article. TEST the system for normal operation after the repair.
Clunk - occurs when shifting from PARK or between REVERSE and DRIVE	Powertrain mount(s)	CHECK the powertrain/drivetrain mounts for damage. INSTALL new mounts as necessary. For engine, REFER to <u>ENGINE - 3.5L</u> article. For transaxle, REFER to <u>AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50</u> article. TEST the system for normal operation after the repair.
-	Idle speed is too high	CHECK for the correct idle speed.
Accessory drive bearing hoot - occurs at idle or high idle in cold temperatures of approximately 4° C (40°F) or colder at the first start of the day	Accessory drive idler or tensioner pulley bearing is experiencing stick/slip between ball bearings and the bearing race	- CARRY OUT the engine cold soak procedure. REFER to <u>NOISE, VIBRATION AND HARSHNESS</u> article. - PLACE the EngineEAR probe directly on the idler/tensioner center post or bolt to verify which bearing is making the noise. INSTALL new parts as necessary. REFER to <u>ACCESSORY DRIVE</u> article. TEST the system for normal operation after the repair.
Accessory drive belt noise, squeal or chirping	- Defective/worn or incorrect accessory drive belt - Misaligned pulley(s) - Pulley runout - Damaged or worn accessory drive component or idler - Fluid contamination of the accessory drive belt or pulleys - Damaged or worn accessory drive belt tensioner - Damaged pulley grooves	CARRY OUT the Engine Accessory Test. REFER to <u>NOISE, VIBRATION AND HARSHNESS</u> article. INSPECT components and INSTALL new parts as necessary. REFER to <u>ACCESSORY DRIVE</u> article. TEST the system for normal operation after the repair.
Clunking noise	Coolant pump has excessive end play or	CHECK the coolant pump for excessive end play. INSPECT the coolant pump for imbalance with the drive belt off. INSTALL a new coolant pump as

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	imbalance	necessary. REFER to <u>ENGINE COOLING</u> article. TEST the system for normal operation after the repair.
Whine or moaning noise	Air intake system	CHECK the air cleaner and ducts for correct fit. INSPECT the air intake system for leaks or damage. REPAIR as necessary. TEST the system for normal operation after the repair.
Whistling noise - normally accompanied with poor idle condition	Air intake system	CHECK the air intake ducts, air cleaner, throttle body and vacuum hoses for leaks and correct fit. REPAIR or ADJUST as necessary. TEST the system for normal operation after the repair.
Hissing noise - occurs during idle or high idle that is apparent with the hood open	Vacuum leak	USE the Ultrasonic Leak Detector/EngineEAR to locate the source. SCAN the air intake system from the inlet to each cylinder intake port. DISCARD the leaking parts, and INSTALL a new component. TEST the system for normal operation after the repair.
-	Vehicles with a plastic intake manifold	Acceptable condition. Some plastic manifolds exhibit this noise, which is the effect of the plastic manifold.
Grinding noise - occurs during engine cranking	- Incorrect starter motor mounting - Starter motor - Incorrect starter motor drive engagement	- INSPECT the starter motor for correct mounting. REPAIR as necessary. REFER to <u>STARTING SYSTEM</u> article. TEST the system for normal operation after the repair. - CHECK the starter motor. INSTALL a new starter motor as necessary. REFER to <u>STARTING SYSTEM</u> article. TEST the system for normal operation after the repair. - INSPECT the starter motor drive for wear or damage. INSTALL a new starter motor as necessary. REFER to <u>STARTING SYSTEM</u> article. TEST the system for normal operation after the repair. - INSPECT the flexplate for wear or damage. INSTALL a new flexplate as necessary. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
Engine noise,		CHECK for obvious signs of damage or

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front of engine - knocking noise from lower front of engine	Damaged or separated crankshaft pulley/damper	wobble during operation. INSTALL new as necessary. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
Engine noise, front of engine - ticking, tapping or rattling noise from the front of the engine	Timing drive components	- REMOVE the accessory drive belt. REFER to <u>ACCESSORY DRIVE</u> article. - USE the EngineEAR to isolate the noise to the engine front cover. - REMOVE the engine front cover and INSPECT the timing drive components. INSTALL new parts as necessary. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
- Engine noise, upper end - ticking noise near the fuel rail and intake manifold - Engine noise, upper end - occurs mostly with a warm engine at light/medium acceleration - Engine noise, upper end - rattling noise from the valve train. Worse when the engine is cold	- Fuel rail clip - Fuel injector - Worn or damaged spark plugs - Carbon accumulation in combustion chamber - Low oil level - Thin or diluted oil	- CHECK for loose or damaged fuel rail clip (s). REPAIR as necessary. TEST the system for normal operation after the repair. - USE the EngineEAR to isolate the noisy injector(s). INSTALL a new injector(s) as necessary. REFER to <u>FUEL CHARGING AND CONTROLS - 3.5L</u> article. TEST the system for normal operation after the repair. - REMOVE the spark plugs. INSPECT and INSTALL new as necessary. REFER to <u>ENGINE IGNITION - 3.5L</u> article. TEST the system for normal operation after the repair. - Bore scope the cylinder. ELIMINATE carbon buildup. TEST the system for normal operation after the repair. - CHECK the oil level. FILL as necessary. - INSPECT the oil for contamination. If the oil is contaminated, CHECK for the source. REPAIR as necessary. CHANGE the oil and filter. TEST the system for normal

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	• Low oil pressure • Worn valve train components • Worn valve guides • Excessive runout of the valve seats on the valve face	operation after the repair. • CARRY OUT an oil pressure test. If not within specifications, REMOVE the engine oil pan. REFER to <u>ENGINE - 3.5L</u> article. INSPECT for a blocked oil pick up tube. TEST the system for normal operation after the repair. • CARRY OUT the Valve Train Analysis Component Test. INSTALL new parts as necessary. TEST the system for normal operation after the repair. • CARRY OUT the Valve Guide Inner Diameter procedure. • CARRY OUT the Valve Seat Inspection procedure.
• Engine noise, upper end - pinging noise	• Gasoline octane too low • Knock Sensor (KS) operation • Incorrect spark timing • High operating temperature • Spark plug • Catalytic converter	• VERIFY with customer the type of gasoline used. CORRECT as necessary. TEST the system for normal operation after the repair. • CHECK the KS. INSTALL a new KS as necessary. REFER to <u>ELECTRONIC ENGINE CONTROLS</u> article. TEST the system for normal operation after the repair. • CHECK the spark timing. REPAIR as necessary. TEST the system for normal operation after the repair. • INSPECT the cooling system for leaks. CHECK the coolant level. REFILL as necessary. CHECK the coolant for the correct mix ratio. DRAIN and REFILL as needed. VERIFY the engine operating temperature is within specifications. REPAIR as necessary. TEST the system for normal operation after the repair. • CHECK the spark plugs. INSTALL new spark plugs as necessary. TEST the system for normal operation after the repair. • Acceptable noise.
• Engine noise, lower end - ticking or		• USE the EngineEAR to verify the oil pump as the source of the noise at low RPM.

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knocking noise near the oil filter adapter	• Oil pump	REPAIR as necessary. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
• Engine noise, lower end - light knocking noise, also described as piston slap. Noise is most noticeable when the engine is cold with light to medium acceleration. The noise disappears as the engine warms	• Excessive clearance between the piston and the cylinder wall	• INSTALL a new short block. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
• Engine noise, lower end - light double knock or sharp rap sound. Occurs mostly with a warm engine at idle or low speeds in drive. Increases in relation to engine load. Associated with a poor lubrication history	• Excessive clearance between the piston and the piston pin	• INSTALL a new short block. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
• Engine noise, lower end - light knocking noise. The noise is most noticeable when the engine is warm. The noise tends to decrease when the vehicle is coasting or in NEUTRAL	• Excessive clearance between the connecting rod bearings and the crankshaft	• INSTALL a new short block. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
• Engine noise, lower end - deep knocking noise. The noise is most noticeable when the engine is	• Worn or damaged crankshaft main	• INSTALL a new short block. REFER to <u>ENGINE - 3.5L</u> article. TEST the system

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warm, at lower RPM and under a light load and then at float	bearings	for normal operation after the repair.
Engine noise, rear of engine - knocking noise at rear of engine	Damaged flywheel/flexplate	CARRY OUT the Flexplate Inspection procedure.
Engine vibration - vibration felt at all times	Excessive engine pulley runout	CARRY OUT the Engine Accessory Test. REFER to <u>NOISE, VIBRATION AND HARSHNESS</u> article. INSTALL a new engine pulley as necessary. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
-	Damaged or worn accessory component	CARRY OUT the Engine Accessory Test. REPAIR or INSTALL a new component as necessary. TEST the system for normal operation after the repair.
Engine vibration - at idle, a low-frequency vibration (5-20 Hz) or mild shake that is felt through the seat/floorpan	Cylinder misfire	Using the scan tool, CARRY OUT the cylinder power balance and the relative compression test. REPAIR as necessary. REFER to <u>ENGINE - 3.5L</u> article. TEST the system for normal operation after the repair.
Engine vibration - is felt with increases and decreases in engine RPM	- Engine or torque converter out of balance - Powertrain mount(s)	- VERIFY the torque converter-to-crankshaft pilot clearance is correct. REPAIR as necessary. RE-INDEX the torque converter on the flexplate by 120 degrees for a 3-bolt converter or 180 degrees for a 4-bolt converter. REFER to <u>AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50</u> article for the transaxle. TEST the system for normal operation after the repair. - CHECK the powertrain mounts for damage. INSTALL new mounts as necessary. For engine, REFER to <u>ENGINE - 3.5L</u> article. For the transaxle, REFER to <u>AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50</u> article. TEST the system for normal operation after the repair.
-	Engine or transmission	INSPECT the powertrain/drivetrain for correct clearances. REPAIR as necessary.

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	grounded to chassis	TEST the system for normal operation after the repair.
Engine vibration - increases intensity as the engine RPM is increased	Engine out-of-balance	CARRY OUT the Neutral Engine Run-Up (NERU) Test. REFER to <u>NOISE, VIBRATION AND HARSHNESS</u> article. ROTATE the torque converter, 120 degrees for 3-bolt or 180 degrees for 4-bolt. INSPECT the torque converter pilot outer diameter-to-crankshaft pilot inner diameter. REPAIR as necessary. REFER to <u>AUTOMATIC TRANSAXLE/TRANSMISSION - 6F50</u> article. TEST the system for normal operation after the repair.
Engine vibration - mostly at coast/neutral coast. Condition improves with vehicle acceleration	Combustion instability	CHECK the ignition system. INSTALL new components as necessary. REFER to <u>ENGINE IGNITION - 3.5L</u> article. TEST the system for normal operation after the repair.
Engine vibration or shudder - occurs with light to medium acceleration above 56 km/h (35 mph)	Worn or damaged spark plugs	INSPECT the spark plugs for cracks, high resistance or broken insulators. INSTALL a new spark plug(s) as necessary. REFER to <u>ENGINE IGNITION - 3.5L</u> article. TEST the system for normal operation after the repair.
- -	- Plugged fuel injector - Contaminated fuel	- REPAIR or INSTALL a new injector as necessary. REFER to <u>FUEL CHARGING AND CONTROLS - 3.5L</u> article. TEST the system for normal operation after the repair. - INSPECT the fuel for contamination. DRAIN the fuel system and refill. REFER to <u>FUEL SYSTEM - GENERAL INFORMATION</u> article.

Component Tests

The following component tests are used to diagnose engine concerns.

Engine Oil Leaks

NOTE: When diagnosing engine oil leaks, the source of the leak must be positively identified prior to repair. If the vehicle is driven extensively between adding the

fluorescent additive and performing the leak test, fan air or wind can spread the leaking oil and make identifying the location of the leak difficult.

Prior to carrying out this procedure, clean the cylinder block, cylinder heads, valve covers, oil pan and flywheel with a suitable solvent to remove all traces of oil.

Engine Oil Leaks - Fluorescent Oil Additive Method

Use the 12 Volt Master UV Diagnostic Inspection Kit to carry out the following procedure for oil leak diagnosis.

1. Add 29.6 ml (1 oz) of gasoline engine oil dye to a minimum of 0.47L (1/2 qt) and a maximum of 0.95L (1 qt) engine oil and fill through the engine oil fill. Thoroughly premix the gasoline engine oil dye or it will not have enough time to reach the crankcase, oil galleries and seal surfaces during this particular 15 minute test. The additive must be mixed well with oil and added through the oil fill. Check the level on the oil level indicator to determine what amount of oil to premix. If it is in the middle of the crosshatch area or below the full mark, use 0.95L (1 qt). If it is at the full mark, use 0.47L (1/2 qt).
2. Run the engine for 15 minutes. Stop the engine and inspect all seal and gasket areas for leaks using the 12 Volt Master UV Diagnostic Inspection Kit. A clear bright yellow or orange area will identify the leak. For extremely small leaks, several hours may be required for the leak to appear.
3. At the end of test, make sure the oil level is within the upper and lower oil indicator marks. Remove oil as necessary if it registers above the full mark.

Leakage Points - Underhood

Examine the following areas for oil leakage:

- Valve cover gaskets
- Cylinder head gaskets
- Oil cooler, if equipped
- Oil filter adapter
- Engine front cover
- Oil filter adapter and filter body
- Oil level indicator tube connection
- Engine Oil Pressure (EOP) switch

Leakage Points - Under Engine, With Vehicle on Hoist

Examine the following areas for oil leakage:

- Oil pan gaskets
- Oil pan sealer
- Engine front cover gasket

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- Crankshaft front seal
- Crankshaft rear oil seal
- Oil filter adapter and filter body
- Oil cooler, if equipped

Leakage Points - With Transmission and Flywheel Removed

Examine the following areas for oil leakage:

- Crankshaft rear oil seal
- Rear main bearing cap parting line
- Flexplate mounting bolt holes (with flexplate installed)
- Pipe plugs at the end of oil passages

Oil leaks at crimped seams in sheet metal parts and cracks in cast or stamped parts can be detected when using the dye method.

Compression Test

1. Make sure the oil in the crankcase is of the correct viscosity and at the correct level and that the battery is correctly charged. Operate the vehicle until the engine is at normal operating temperature. Turn the ignition switch to the OFF position, then remove all the spark plugs.
2. Set the throttle plates in the wide-open position.
3. Install a compression gauge such as the Compression Tester in the No. 1 cylinder.
4. Install an auxiliary starter switch in the starting circuit. With the ignition switch in the OFF position, and using the auxiliary starter switch, crank the engine a minimum of 5 compression strokes and record the highest reading. Note the approximate number of compression strokes necessary to obtain the highest reading.
5. Repeat the test on each cylinder, cranking the engine approximately the same number of compression strokes.

Compression Test - Test Results

The indicated compression pressures are considered within specification if the lowest reading cylinder is at least 75% of the highest reading. Refer to the COMPRESSION PRESSURE LIMIT CHART.

COMPRESSION PRESSURE LIMIT CHART

Maximum Pressure	Minimum Pressure	Maximum Pressure	Minimum Pressure	Maximum Pressure	Minimum Pressure	Maximum Pressure	Minimum Pressure
924 kPa (134 psi)	696 kPa (101 psi)	1,131 kPa (164 psi)	848 kPa (123 psi)	1,338 kPa (194 psi)	1,000 kPa (146 psi)	1,544 kPa (224 psi)	1,158 kPa (168 psi)
938 kPa (136 psi)	703 kPa (102 psi)	1,145 kPa (166 psi)	855 kPa (124 psi)	1,351 kPa (196 psi)	1,014 kPa (147 psi)	1,558 kPa (226 psi)	1,165 kPa (169 psi)

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952 kPa (138 psi)	717 kPa (104 psi)	1,158 kPa (168 psi)	869 kPa (126 psi)	1,365 kPa (198 psi)	1,020 kPa (148 psi)	1,572 kPa (228 psi)	1,179 kPa (171 psi)
965 kPa (140 psi)	724 kPa (106 psi)	1,172 kPa (170 psi)	876 kPa (127 psi)	1,379 kPa (200 psi)	1,034 kPa (150 psi)	1,586 kPa (230 psi)	1,186 kPa (172 psi)
979 kPa (142 psi)	738 kPa (107 psi)	1,186 kPa (172 psi)	889 kPa (129 psi)	1,303 kPa (202 psi)	1,041 kPa (151 psi)	1,600 kPa (232 psi)	1,200 kPa (174 psi)
933 kPa (144 psi)	745 kPa (109 psi)	1,200 kPa (174 psi)	903 kPa (131 psi)	1,407 kPa (204 psi)	1,055 kPa (153 psi)	1,055 kPa (153 psi)	1,207 kPa (175 psi)
1,007 kPa (146 psi)	758 kPa (110 psi)	1,214 kPa (176 psi)	910 kPa (132 psi)	1,420 kPa (206 psi)	1,062 kPa (154 psi)	1,627 kPa (154 psi)	1,220 kPa (177 psi)
1,020 kPa (148 psi)	765 kPa (111 psi)	1,227 kPa (178 psi)	917 kPa (133 psi)	1,434 kPa (208 psi)	1,075 kPa (156 psi)	1,641 kPa (238 psi)	1,227 kPa (178 psi)
1,034 kPa (150 psi)	779 kPa (113 psi)	1,241 kPa (180 psi)	931 kPa (135 psi)	1,448 kPa (210 psi)	1,083 kPa (157 psi)	1,655 kPa (240 psi)	1,241 kPa (180 psi)
1,048 kPa (152 psi)	786 kPa (114 psi)	1,255 kPa (182 psi)	936 kPa (136 psi)	1,462 kPa (212 psi)	1,089 kPa (158 psi)	1,669 kPa (242 psi)	1,248 kPa (181 psi)
1,062 kPa (154 psi)	793 kPa (115 psi)	1,269 kPa (184 psi)	952 kPa (138 psi)	1,476 kPa (214 psi)	1,103 kPa (160 psi)	1,682 kPa (244 psi)	1,262 kPa (183 psi)
1,076 kPa (156 psi)	807 kPa (117 psi)	1,282 kPa (186 psi)	965 kPa (140 psi)	1,489 kPa (216 psi)	1,117 kPa (162 psi)	1,696 kPa (246 psi)	1,269 kPa (184 psi)
1,089 kPa (158 psi)	814 kPa (118 psi)	1,296 kPa (188 psi)	972 kPa (141 psi)	1,503 kPa (218 psi)	1,124 kPa (163 psi)	1,710 kPa (248 psi)	1,202 kPa (186 psi)
1,103 kPa (160 psi)	827 kPa (120 psi)	1,310 kPa (190 psi)	979 kPa (142 psi)	1,517 kPa (220 psi)	1,138 kPa (165 psi)	1,724 kPa (250 psi)	1,289 kPa (187 psi)
1,110 kPa (161 psi)	834 kPa (121 psi)	1,324 kPa (192 psi)	993 kPa (144 psi)	1,631 kPa (222 psi)	1,145 kPa (166 psi)	-	-

If one or more cylinders reads low, squirt approximately one tablespoon of engine oil meeting Ford specification on top of the pistons in the low-reading cylinders. Repeat the compression pressure check on these cylinders.

Compression Test - Interpreting Compression Readings

1. If compression improves considerably, piston rings are worn or damaged.
2. If compression does not improve, valves are sticking or not seating correctly.
3. If 2 adjacent cylinders indicate low compression pressures and squirting oil on each piston does not increase compression, the head gasket may be leaking between cylinders. Engine oil or coolant in cylinders could result from this condition.

Use the Compression Pressure Limit Chart when checking cylinder compression so that the lowest reading is within 75% of the highest reading.

Cylinder Leakage Detection

When a cylinder produces a low reading, use of the Engine Cylinder Leak Detection/Air Pressurization Kit will

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be helpful in pinpointing the exact cause.

The leakage detector is inserted in the spark plug hole, the piston is brought up to Top Dead Center (TDC) on the compression stroke, and compressed air is admitted.

Once the combustion chamber is pressurized, a special gauge included in the kit will read the percentage of leakage. Leakage exceeding 20% is excessive.

While the air pressure is retained in the cylinder, listen for the hiss of escaping air. A leak at the intake valve will be heard in the Throttle Body (TB). A leak at the exhaust valve can be heard at the tailpipe. Leakage past the piston rings will be audible at the PCV connection. If air is passing through a blown head gasket to an adjacent cylinder, the noise will be evident at the spark plug hole of the cylinder into which the air is leaking. Cracks in the cylinder block or gasket leakage into the cooling system may be detected by a stream of bubbles in the radiator.

Excessive Engine Oil Consumption

Nearly all engines consume oil, which is essential for normal lubrication of the cylinder bore walls and pistons and rings. Determining the level of oil consumption may require testing by recording how much oil is being added over a given set of miles.

Customer driving habits greatly influence oil consumption. Mileage accumulated during towing or heavy loading generates extra heat. Frequent short trips, stop-and-go type traffic or extensive idling, prevent the engine from reaching normal operating temperature. This prevents component clearances from reaching specified operating ranges.

The following diagnostic procedure may be utilized to determine internal oil consumption. Make sure that the concern is related to internal oil consumption, and not external leakage, which also consumes oil. Verify there are no leaks before carrying out the test. Once verified, the rate of internal oil consumption can be tested.

A new engine may require extra oil in the early stages of operation. Internal piston-to-bore clearances and sealing characteristics improve as the engine breaks in. Engines are designed for close tolerances and do not require break-in oils or additives. Use the oil specified in the Owner's Literature. Ambient temperatures may determine the oil viscosity specification. Verify that the correct oil is being used for the vehicle in the geographic region in which it is driven.

Basic Pre-checks

1. For persistent complaints of oil consumption, interview the customer to determine the oil consumption characteristics. If possible, determine the brand and grade of oil currently in the oil pan. Look at the oil filter or oil-change station tags to determine if Ford-recommended maintenance schedules have been followed. Make sure that the oil has been changed at the specified mileage intervals. If vehicle mileage is past the first recommended drain interval, the OEM production filter should have been changed.
2. Ask how the most current mileage was accumulated. That is, determine whether the vehicle was driven under the following conditions:
 - Extended idling or curbside engine operation
 - Stop-and-go traffic or taxi operation

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- Towing a trailer or vehicle loaded heavily
 - Frequent short trips (engine not up to normal operating temperature)
 - Excessive throttling or high engine-RPM driving
3. Verify that there are no external leaks. If necessary, review the diagnostic procedure under **ENGINE OIL LEAKS** in Diagnosis and Testing .
 4. Inspect the crankcase ventilation system for:
 - disconnected hoses at the valve cover or TB.
 - loose or missing valve cover fill cap.
 - missing or incorrectly seated engine oil level indicator.
 - incorrect or dirty PCV valve.
 - a PCV valve grommet unseated in the valve cover (if so equipped).
 5. Inspect for signs of sludge. Sludge affects PCV performance and can plug or restrict cylinder head drainback wells. It can also increase oil pressure by restricting passages and reducing the drainback capability of piston oil control rings. Sludge can result from either excessive water ingestion in the crankcase or operation at extremely high crankcase temperatures.
 6. Inspect the air filter for dirt, sludge or damage. A hole in the filter element will allow unfiltered air to bypass into the air induction system. This can cause premature internal wear (engine dusting), allowing oil to escape past rings, pistons, valves and guides.
 7. If the engine is hot or was recently shut down, wait at least 5 minutes to allow the oil to drain back. Ask the customer if this requirement has been followed. Adding oil without this wait period can cause an overfill condition, leading to excessive oil consumption and foaming, which may cause engine damage.
 8. Make sure the oil level indicator (dipstick) is correctly and fully seated in the indicator tube. Remove the oil level indicator and record the oil level.

Detailed Pre-checks

1. Check the thermostat opening temperature to make sure that the cooling system is operating at the specified temperature. If it is low, internal engine parts are not running at specified internal operating clearances.
2. Verify the spark plugs are not oil saturated. Oil leaking into one or more cylinders will appear as an oil soaked condition on the plug. If a plug is saturated, a compression check may be necessary at the conclusion of the oil consumption test.

Oil Consumption Test

Once all of the previous conditions are met, carry out an oil consumption test.

1. Drain the engine oil and remove the oil filter. Install a new manufacturer-specified oil filter. Make sure the vehicle is positioned on a level surface. Refill the oil pan to a level **one liter (quart) less** than the specified fill level, using manufacturer-specified oil.
2. Run the engine for 3 minutes (if hot) or 10 minutes (if cold). Allow for a minimum 5-minute drainback period and then record the oil level shown on the oil level indicator. Place a mark on the backside of the oil level indicator noting the oil level location.

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3. Add the final one liter (quart) to complete the normal oil fill. Restart the engine and allow it to idle for 2 minutes. Shut the engine down.
4. After a 5-minute drainback period, record the location of the oil level again. Mark the oil level indicator with the new oil level location. (Note: Both marks should be very close to the MIN-MAX upper and lower limits or the upper and lower holes on the oil level indicator. These marks will exactly measure the engine's use of oil, with a one quart differential between the new marks.) Demonstrate to the customer that the factory-calibrated marks on the oil level indicator are where the oil should fall after an oil change with the specified fill amount. Explain however, that this may vary slightly between MIN-MAX or the upper and lower holes on the oil level indicator.
5. Record the vehicle mileage.
6. Advise the customer that oil level indicator readings must be taken every 320 km (200 mi) or weekly, using the revised marks as drawn. Remind the customer that the engine needs a minimum 5-minute drainback for an accurate reading and that the oil level indicator must be firmly seated in the tube prior to taking the reading.
7. When the subsequent indicator readings demonstrate a full liter (quart) has been used, record the vehicle mileage. The mileage driven between the 2 readings should not be less than 2,414 km (1,500 mi). The drive cycle the vehicle has been operated under must be considered when making this calculation. It may be necessary to have the customer bring the vehicle in for a periodic oil level indicator reading to closely monitor oil usage.

Post Checks, Evaluation and Corrective Action

1. If test results indicate excessive oil consumption, carry out a cylinder compression test. The cylinder compression test should be carried out with a fully charged battery and all spark plugs removed. See the **COMPRESSION PRESSURE LIMIT CHART** for pressure range limits.
2. Compression should be consistent across all cylinders. Refer to **COMPRESSION TEST**. If compression tested within the specifications found in this service information, the excessive oil consumption may be due to wear on the valve guides, valves or valve seals.
3. A cylinder leak detection test can be carried out using an Engine Cylinder Leak Detection/Air Pressurization Kit. This can help identify valves, piston rings, or worn valve guides/valve stems, inoperative valve stem seals or other related areas as the source of oil consumption.

NOTE: **An oil-soaked appearance on the porcelain tips of the spark plugs also indicates excessive oil use. A typical engine with normal oil consumption will exhibit a light tan to brown appearance. See Spark Plug Analysis in this service information for details. A single or adjoining, multiple cylinder leak can be traced by viewing the tips.**

4. If an internal engine part is isolated as the root cause, determine if the repair will exceed cost limits and proceed with a repair strategy as required.
5. Once corrective action to engine is complete and verifying that all pre-check items were eliminated in the original diagnosis, repeat the Oil Consumption Test as described above and verify consumption results.

Intake Manifold Vacuum Test

Bring the engine to normal operating temperature. Connect the Vacuum/Pressure Tester to the intake manifold. Run the engine at the specified idle speed.

The vacuum gauge should read between 51-74 kPa (15-22 in-Hg) depending upon the engine condition and the altitude at which the test is conducted. Subtract 4.0193 kPa (1 in-Hg) from the specified reading for every 304.8 m (1,000 ft) of elevation above sea level.

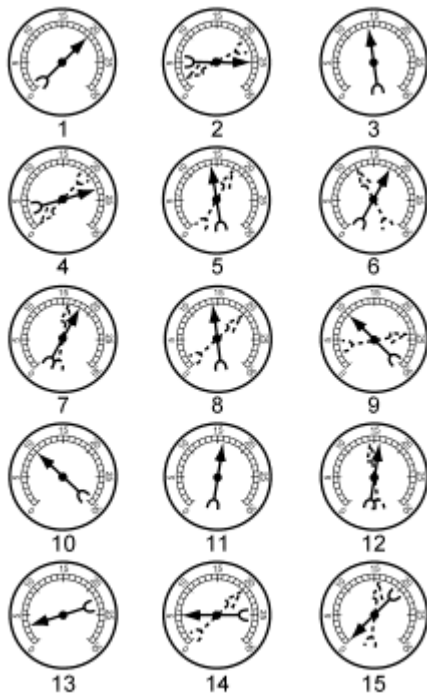
The reading should be steady. If necessary, adjust the gauge damper control (where used) if the needle is fluttering rapidly. Adjust the damper until the needle moves easily without excessive flutter.

Intake Manifold Vacuum Test - Interpreting Vacuum Gauge Readings

A careful study of the vacuum gauge reading while the engine is idling will help pinpoint trouble areas. Always conduct other appropriate tests before arriving at a final diagnostic decision. Vacuum gauge readings, although helpful, must be interpreted carefully.

Most vacuum gauges have a normal band indicated on the gauge face.

The following are potential gauge readings. Some are normal; others should be investigated further.



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Fig. 1: Potential Vacuum Gauge Readings Chart
Courtesy of FORD MOTOR CO.

1. NORMAL READING: Needle between 51-74 kPa (15-22 in-Hg) and holding steady.
2. NORMAL READING DURING RAPID ACCELERATION AND DECELERATION: When the engine is rapidly accelerated (dotted needle), the needle will drop to a low reading (not to zero). When the

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throttle is suddenly released, the needle will snap back up to a higher than normal figure.

3. **NORMAL FOR HIGH-LIFT CAMSHAFT WITH LARGE OVERLAP:** The needle will register as low as 51 kPa (15 in-Hg) but will be relatively steady. Some oscillation is normal.
4. **WORN RINGS OR DILUTED OIL:** When the engine is accelerated (dotted needle), the needle drops to 0 kPa (0 in-Hg). Upon deceleration, the needle runs slightly above 74 kPa (22 in-Hg).
5. **STICKING VALVES:** When the needle (dotted) remains steady at a normal vacuum but occasionally flicks (sharp, fast movement) down and back about 13 kPa (4 in-Hg), one or more valves may be sticking.
6. **BURNED OR WARPED VALVES:** A regular, evenly-spaced, downscale flicking of the needle indicates one or more burned or warped valves. Insufficient valve clearance will also cause this reaction.
7. **POOR VALVE SEATING:** A small but regular downscale flicking can mean one or more valves are not seating.
8. **WORN VALVE GUIDES:** When the needle oscillates over about a 13 kPa (4 in-Hg) range at idle speed, the valve guides could be worn. As engine speed increases, the needle will become steady if guides are responsible.
9. **WEAK VALVE SPRINGS:** When the needle oscillation becomes more violent as engine RPM is increased, weak valve springs are indicated. The reading at idle could be relatively steady.
10. **LATE VALVE TIMING:** A steady but low reading could be caused by late valve timing.
11. **IGNITION TIMING RETARDING:** Retarded ignition timing will produce a steady but somewhat low reading.
12. **INSUFFICIENT SPARK PLUG GAP:** When spark plugs are gapped too close, a regular, small pulsation of the needle can occur.
13. **INTAKE LEAK:** A low, steady reading can be caused by an intake manifold or throttle body gasket leak.
14. **BLOWN HEAD GASKET:** A regular drop of fair magnitude can be caused by a blown head gasket or warped cylinder head-to-cylinder block surface.
15. **RESTRICTED EXHAUST SYSTEM:** When the engine is first started and is idled, the reading may be normal, but as the engine RPM is increased, the back pressure caused by a clogged muffler, kinked tailpipe or other concerns will cause the needle to slowly drop to 0 kPa (0 in-Hg). The needle then may slowly rise. Excessive exhaust clogging will cause the needle to drop to a low point even if the engine is only idling.
16. When vacuum leaks are indicated, search out and correct the cause. Excess air leaking into the system will upset the fuel mixture and cause concerns such as rough idle, missing on acceleration or burned valves. If the leak exists in an accessory unit such as the power brake booster, the unit will not function correctly. Always fix vacuum leaks.

Oil Pressure Test

1. Disconnect and remove the oil pressure switch from the engine.
2. Connect the engine Oil Pressure Gauge to the oil pressure switch oil galley port.
3. Run the engine until normal operating temperature is reached.
4. Run the engine at the specified RPM and record the gauge reading.
5. The oil pressure should be within specifications; refer to the specification chart in the appropriate engine section.

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6. If the pressure is not within specification, check the following possible sources:

- Insufficient oil
- Oil leakage
- Worn or damaged oil pump
- Oil pump screen cover and tube
- Excessive main bearing clearance
- Excessive connecting rod bearing clearance
- Chain tensioner leak

Valve Train Analysis

The following component tests are used to diagnose valve train concerns.

Valve Train Analysis - Engine Off, Valve Cover Removed

Check for damaged or severely worn parts and correct assembly. Make sure correct parts are used with the static engine analysis as follows.

Valve Train Analysis - Camshafts and Valve Tappets

- Check for broken or damaged parts.
- Check for loose mounting bolts on camshaft caps.
- Check for worn or damaged valve tappets.

Valve Train Analysis - Valve Springs, Valve Tappets Removed

- Check for broken or damaged parts.

Valve Train Analysis - Valve Spring Retainer and Valve Spring Retainer Keys, Valve Tappets Removed

- Check for correct seating of the valve spring retainer key on the valve stem and in valve spring retainer.
- Check for correct seating on the valve stem.

Valve Train Analysis - Valves and Cylinder Head, Valve Tappets Removed

- Check for plugged oil drain-back holes.
- Check for worn or damaged valve tips.
- Check for missing or damaged valve stem seals or guide-mounted valve stem seal.
- Check for missing or worn valve spring seats.

Valve Train Analysis - Camshaft Lobe Lift

Check the lift of each camshaft lobe in consecutive order and make a note of the readings.

1. Remove the spark plugs. Refer to **ENGINE IGNITION - 3.5L** article.
2. Install the Dial Indicator Gauge with Holding Fixture so the rounded tip of the dial indicator is on top of the camshaft lobe and on the same plane as the valve tappet.
3. Rotate the crankshaft using a breaker bar and socket attached to the crankshaft pulley retainer bolt. Rotate the crankshaft until the base circle of the camshaft lobe is reached.

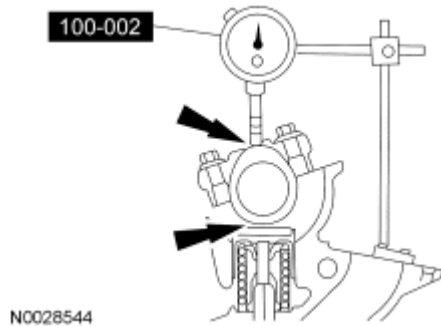


Fig. 2: Checking Camshaft Lobe Lift
Courtesy of FORD MOTOR CO.

4. Zero the dial indicator. Continue to rotate the crankshaft until the high-lift point of the camshaft lobe is in the fully-raised position (highest indicator reading).
5. To check the accuracy of the original dial indicator reading, continue to rotate crankshaft until the base circle is reached. The dial indicator reading should be zero. If zero reading is not obtained, repeat Steps 1 through 6.
6. If the lift on any lobe is below specified service limits, install a new camshaft and camshaft valve tappets.
7. Install the spark plugs. Refer to **ENGINE IGNITION - 3.5L** article.

GENERAL PROCEDURES

SPROCKETS

1. Inspect the sprockets for cracks and worn or chipped teeth.

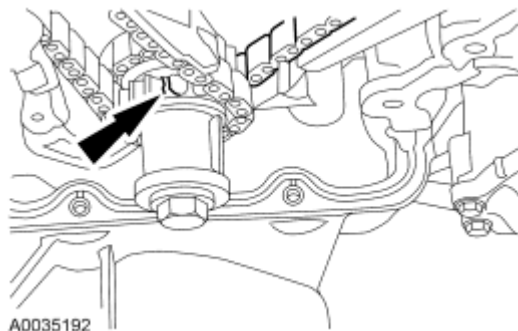
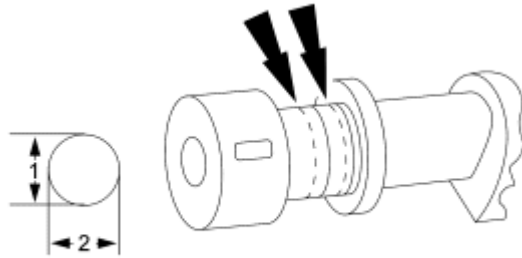


Fig. 3: Inspecting Sprockets For Damage
Courtesy of FORD MOTOR CO.

CAMSHAFT BEARING JOURNAL DIAMETER

NOTE: Refer to the appropriate service information for the specification.

1. Measure each camshaft journal diameter in 2 directions.



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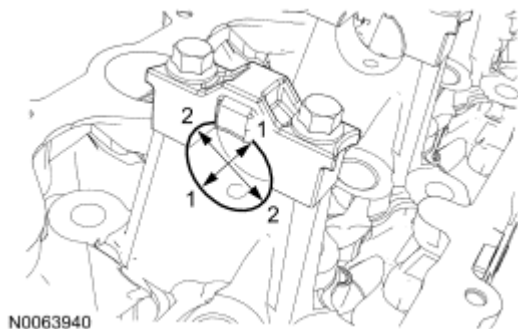
Fig. 4: Locating Areas For Measuring Camshaft Journal Diameter
Courtesy of FORD MOTOR CO.

CAMSHAFT JOURNAL TO BEARING CLEARANCE - OHC ENGINES

NOTE: Refer to the appropriate service information for the specification.

NOTE: The camshaft journals must meet specifications before checking camshaft journal clearance.

1. Measure each camshaft bearing in 2 directions.
 - Subtract the camshaft journal diameter from the camshaft bearing diameter.



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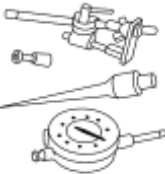
Fig. 5: Measuring Camshaft Bearing In 2 Directions
Courtesy of FORD MOTOR CO.

CAMSHAFT END PLAY - OHC ENGINES**SPECIAL TOOLS**

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Illustration	Tool Name	Tool Number
 ST1214-A	Dial Indicator Gauge with Holding Fixture	100-002 (TOOL-4201-C) or equivalent

NOTE: Refer to the appropriate service information for the specification.

1. Using the Dial Indicator Gauge with Holding Fixture, measure the camshaft end play.
2. Position the camshaft to the rear of the cylinder head.
3. Zero the Dial Indicator Gauge.
4. Move the camshaft to the front of the cylinder head. Note and record the camshaft end play.
 - If camshaft end play exceeds specifications, install a new camshaft and recheck end play.
 - If camshaft end play exceeds specification after camshaft installation, install a new cylinder head.

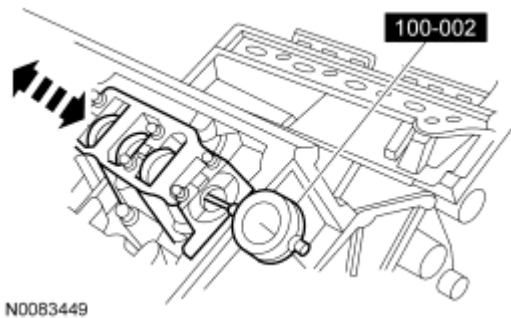
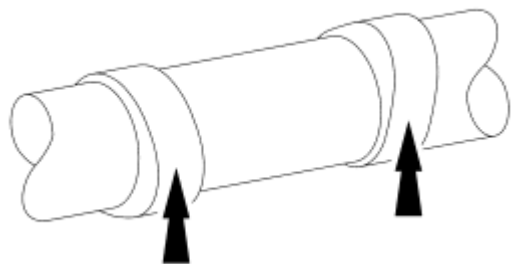


Fig. 6: Recording Camshaft End Play
Courtesy of FORD MOTOR CO.

CAMSHAFT SURFACE INSPECTION

1. Inspect camshaft lobes for pitting or damage in the contact area. Minor pitting is acceptable outside the contact area.



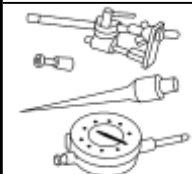
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Fig. 7: Inspecting Camshaft Lobes
Courtesy of FORD MOTOR CO.

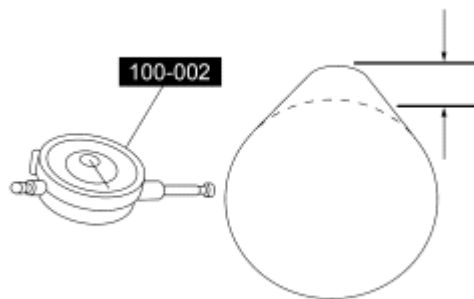
CAMSHAFT LOBE LIFT

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1214-A	Dial Indicator Gauge with Holding Fixture	100-002 (TOOL-4201-C) or equivalent

NOTE: Refer to the appropriate service information for the specification.

1. Use the Dial Indicator Gauge with Holding Fixture to measure camshaft intake/exhaust lobe lift.
 - Rotate the camshaft and subtract the lowest indicator reading from the highest indicator reading to figure the camshaft lobe lift.

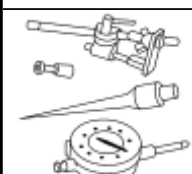


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Fig. 8: Using Dial Indicator Gauge With Holding Fixture To Measure Camshaft Intake/Exhaust Lobe Lift
Courtesy of FORD MOTOR CO.

CAMSHAFT RUNOUT

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1214-A	Dial Indicator Gauge with Holding Fixture	100-002 (TOOL-4201-C) or equivalent

NOTE: Refer to the appropriate service information for the specification.

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NOTE: Camshaft journals must be within specifications before checking runout.

1. Using the Dial Indicator Gauge with Holding Fixture, measure the camshaft runout.
 - Rotate the camshaft and subtract the lowest indicator reading from the highest indicator reading.

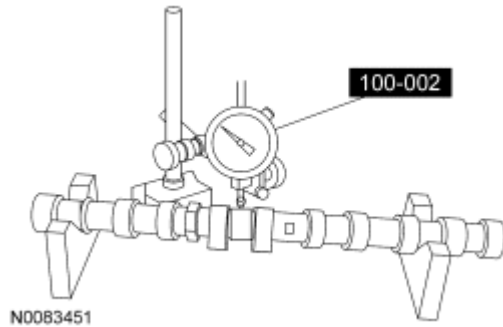
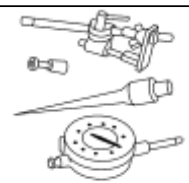


Fig. 9: Measuring Camshaft Runout Using Dial Indicator Gauge With Holding Fixture
Courtesy of FORD MOTOR CO.

CRANKSHAFT END PLAY

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1214-A	Dial Indicator Gauge with Holding Fixture	100-002 (TOOL-4201-C) or equivalent

NOTE: Refer to the appropriate service information for the specification.

1. Install the Dial Indicator Gauge with Holding Fixture.
2. Position the crankshaft to the rear of the cylinder block.
3. Zero the Dial Indicator Gauge.
4. Move the crankshaft to the front of the cylinder block. Note and record the crankshaft end play.

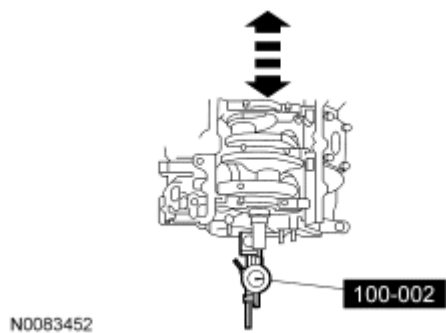


Fig. 10: Recording Crankshaft End Play
Courtesy of FORD MOTOR CO.

VALVE STEM DIAMETER

NOTE: Refer to the appropriate service information for the specification.

1. Measure the diameter of each intake and exhaust valve stem at the points shown in illustration. Verify the diameter is within specification.

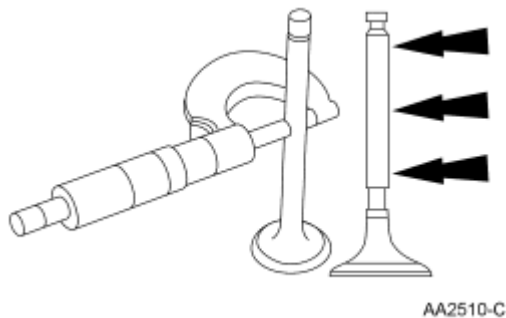


Fig. 11: Measuring Diameter Of Valve Stem
Courtesy of FORD MOTOR CO.

VALVE INSPECTION

1. Inspect the following valve areas:
 1. The end of the stem for grooves or scoring.
 2. The valve face and the edge for pits, grooves or scores.
 3. The valve head for signs of burning, erosion, warpage and cracking.
 4. The valve margin for wear.

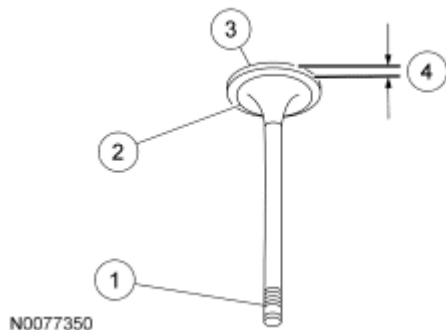


Fig. 12: Inspecting Valves
Courtesy of FORD MOTOR CO.

VALVE GUIDE INNER DIAMETER

NOTE: Refer to the appropriate service information for the specification.

NOTE: Valve guides tend to wear in an hourglass pattern. The ball gauge can be inserted into the combustion chamber side of the valve guide, if necessary.

- 1. Use a ball gauge to determine the inside diameter of the valve guides in 2 directions at the top, middle and bottom of the valve guide.

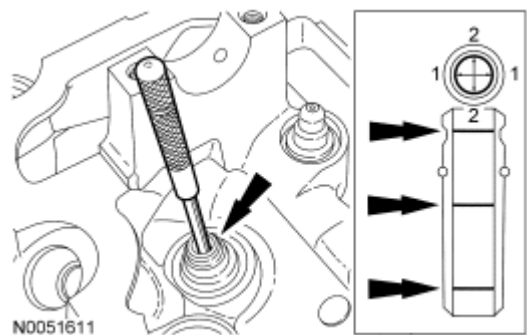


Fig. 13: Measuring Valve Guide Inner Diameter
Courtesy of FORD MOTOR CO.

- 2. Measure the ball gauge with a micrometer.

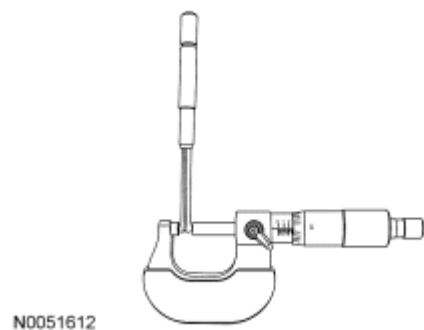


Fig. 14: Measuring Ball Gauge With Micrometer
Courtesy of FORD MOTOR CO.

- 3. If the valve guide is not within specifications, install a new cylinder head assembly.

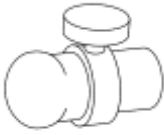
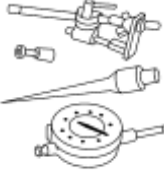
VALVE STEM TO VALVE GUIDE CLEARANCE

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
	Clearance Gauge, Valve Guide	303-004 (TOOL-6505-E) or

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 ST1251-A		equivalent
 ST1214-A	Dial Indicator Gauge with Holding Fixture	100-002 (TOOL-4201-C) or equivalent

NOTE: Refer to the appropriate service information for the specification.

NOTE: The valve stem diameter must be within specifications before checking valve stem-to-valve guide clearance.

NOTE: If necessary, use a magnetic base.

1. Install a Valve Guide Clearance Gauge on the valve stem and install a Dial Indicator Gauge with Holding Fixture. Lower the valve until the clearance gauge contacts the upper surface of the valve guide.

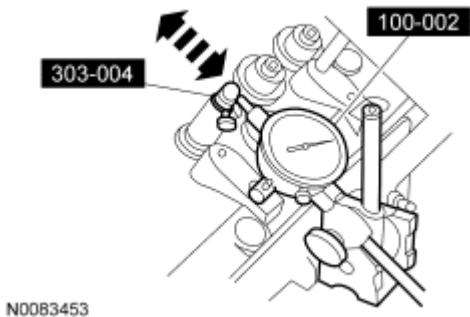


Fig. 15: Installing Valve Guide Clearance Gauge
Courtesy of FORD MOTOR CO.

2. Move the Valve Guide Clearance Gauge toward the Dial Indicator Gauge with Holding Fixture and zero the Dial Indicator Gauge. Move the Valve Guide Clearance Gauge away from the Dial Indicator Gauge with Holding Fixture and note the reading. The reading will be DOUBLE the valve stem-to-valve guide clearance.

VALVE SPRING INSTALLED LENGTH

NOTE: Refer to the appropriate service information for the specification.

1. Measure the installed length of each valve spring.

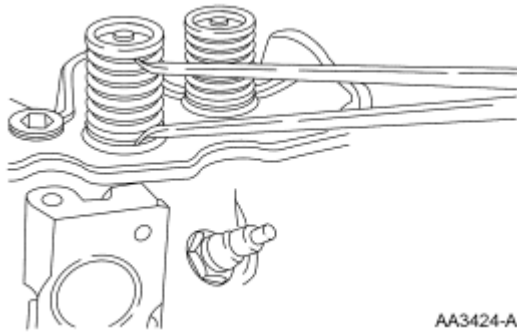


Fig. 16: Measuring Installed Length Of Valve Spring
Courtesy of FORD MOTOR CO.

VALVE SPRING FREE LENGTH

NOTE: Refer to the appropriate service information for the specification.

1. Measure the free length of each valve spring.

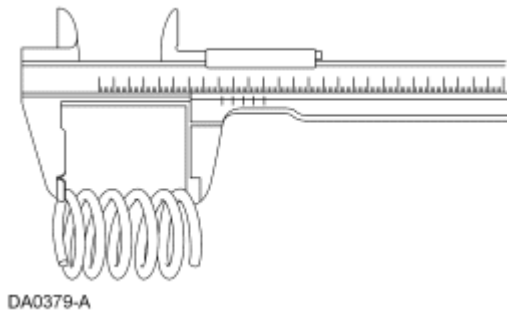


Fig. 17: Measuring Free Length Of Valve Spring
Courtesy of FORD MOTOR CO.

VALVE SPRING SQUARENESS

1. Measure the out-of-square on each valve spring.
 - Turn the valve spring and observe the space between the top of the valve spring and the square.

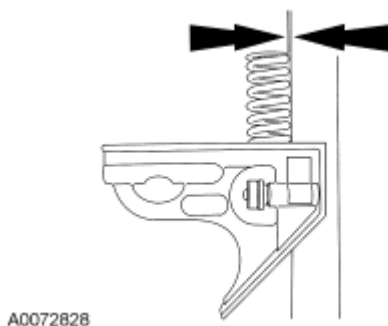


Fig. 18: Measuring Valve Spring Out-Of-Square
Courtesy of FORD MOTOR CO.

VALVE SPRING STRENGTH

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1278-A	Pressure Gauge, Valve/Clutch Spring	303-006 (TOOL-6513-DD) or equivalent

NOTE: Refer to the appropriate service information for the specification.

1. Use the Valve/Clutch Spring Pressure Gauge to check the valve spring for correct strength at the specified valve spring length.

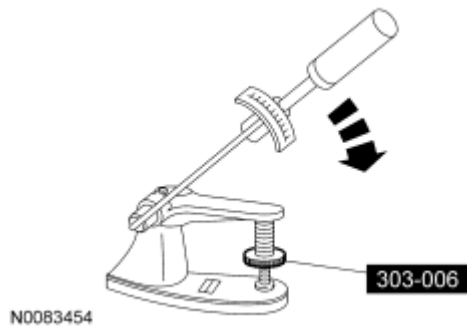


Fig. 19: Using Valve/Clutch Spring Pressure Gauge
Courtesy of FORD MOTOR CO.

VALVE SEAT INSPECTION

Valve and Seat Refacing Measurements

NOTE: Refer to the appropriate service information for the specification.

NOTE: After grinding valves or valve seats, check valve clearance.

1. Check the valve head and seat.
 - Check valve angles.
 - Check margin width.
 - Be sure margin width is within specification.

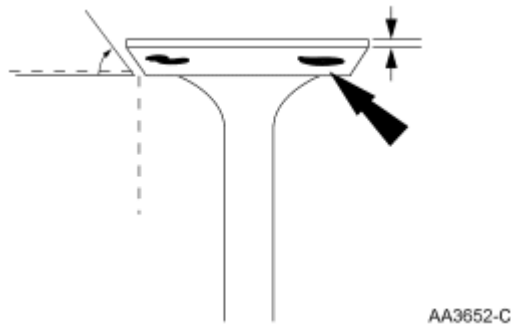


Fig. 20: Checking Valve Head And Seat
Courtesy of FORD MOTOR CO.

2. Inspect for abnormalities on the valve face and seat. Install a new cylinder head assembly if abnormalities are found.

VALVE SEAT WIDTH

NOTE: Refer to the appropriate service information for the specification.

1. Measure the valve seat width. If necessary, grind the valve seat to specification.
 - Measure the intake valve seat width.
 - Measure the exhaust valve seat width.
 - Recheck the valve spring installed length after the seats have been ground, and shim the valve springs as necessary to achieve the correct installed spring length.



Fig. 21: Measuring Valve Seat Width
Courtesy of FORD MOTOR CO.

VALVE SEAT RUNOUT

NOTE: Refer to the appropriate service information for the specification.

1. Use a valve seat runout gauge to check valve seat runout.

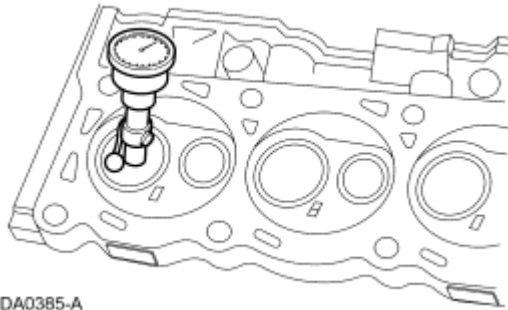


Fig. 22: Measuring Valve Seat Runout
Courtesy of FORD MOTOR CO.

FLEXPLATE INSPECTION

1. Inspect the flexplate for:
 1. any cracks.
 2. worn ring gear teeth.
 3. chipped or cracked ring gear teeth.

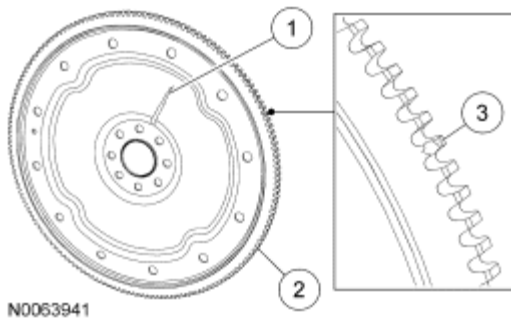


Fig. 23: Inspecting Flexplate
Courtesy of FORD MOTOR CO.

CYLINDER HEAD DISTORTION

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
ST1271-A	Feeler Gauge Set	303-D027 (D81L-4201-A) or equivalent

NOTE: Refer to the appropriate service information for the specification.

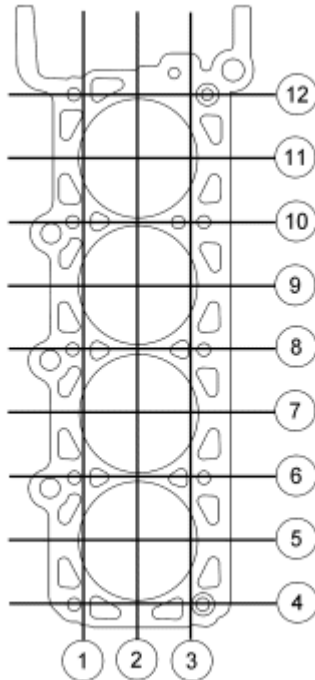
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NOTE: Make sure all cylinder head surfaces are clear of any gasket material, silicone sealant, oil and coolant. The cylinder head surface must be clean and dry before running a flatness check.

NOTE: Use a Straightedge that is calibrated by the manufacturer to be flat within 0.005 mm (0.0002 in) per running foot of length, such as Snap-On® GA438A or equivalent. For example, if the Straightedge is 61 cm (24 in) long, the machined edge must be flat within 0.010 mm (0.0004 in) from end to end.

1. Using a Straightedge and a Feeler Gauge Set, inspect the cylinder head for flatness in the sequence shown in illustration.

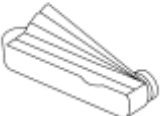


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Fig. 24: Inspecting Cylinder Head For Flatness
Courtesy of FORD MOTOR CO.

CYLINDER BLOCK DISTORTION

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1271-A	Feeler Gauge Set	303-D027 (D81L-4201-A) or equivalent

NOTE: Refer to the appropriate service information for the specification.

NOTE: Use a Straightedge that is calibrated by the manufacturer to be flat within 0.005 mm (0.0002 in) per running foot of length, such as Snap-On® GA438A or equivalent. For example, if the Straightedge is 61 cm (24 in) long, the machined edge must be flat within 0.010 mm (0.0004 in) from end to end.

1. Use a Straightedge and a Feeler Gauge Set to inspect the cylinder block for flatness.

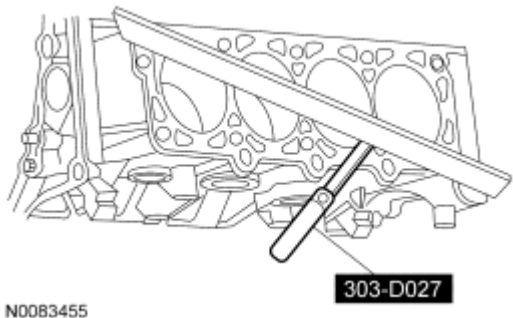
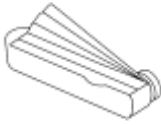


Fig. 25: Inspecting Cylinder Block For Flatness Using Straightedge & Feeler Gauge Set
Courtesy of FORD MOTOR CO.

EXHAUST MANIFOLD CLEANING AND INSPECTION

SPECIAL TOOLS

Illustration	Tool Name	Tool Number
 ST1271-A	Feeler Gauge Set	303-D027 (D81L-4201-A) or equivalent

1. Clean the exhaust manifold using a suitable solvent. Use a plastic scraping tool to clean the gasket sealing surfaces.

NOTE: New exhaust manifold gaskets, studs, nuts and/or bolts must be installed when an exhaust manifold is serviced.

NOTE: Use a Straightedge that is calibrated by the manufacturer to be flat within 0.005 mm (0.0002 in) per running foot of length, such as Snap-On® GA438A or equivalent. For example, if the Straightedge is 61 cm (24 in) long, the machined edge must be flat within 0.010 mm (0.0004 in) from end to end.

2. Using the Straightedge and a Feeler Gauge Set, check the exhaust manifold sealing surface for warpage. If the warpage is greater than 0.76 mm (0.0299 in), install a new exhaust manifold.

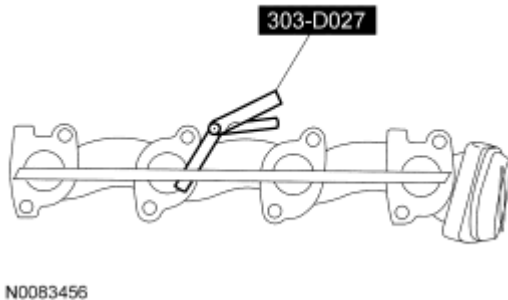


Fig. 26: Checking Exhaust Manifold Sealing Surface For Warpage
Courtesy of FORD MOTOR CO.

SPARK PLUG INSPECTION

1. Inspect the spark plug for a bridged gap.
 - Check for deposit build-up closing the gap between the electrodes. Deposits are caused by oil or carbon fouling.
 - Install a new spark plug.



Fig. 27: Identifying Spark Plug With Bridged Gap
Courtesy of FORD MOTOR CO.

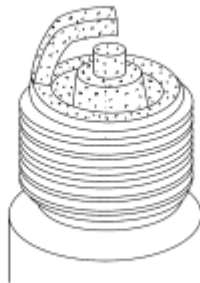
2. Check for oil fouling.
 - Check for wet, black deposits on the insulator shell bore electrodes, caused by excessive oil entering the combustion chamber through worn rings and pistons, excessive valve-to-guide clearance or worn or loose bearings.
 - Correct the oil leak concern.
 - Install a new spark plug.



AB0036-A

Fig. 28: Identifying Oil Fouled Spark Plug
Courtesy of FORD MOTOR CO.

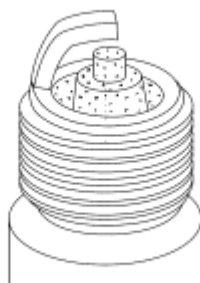
3. Inspect for carbon fouling. Look for black, dry, fluffy carbon deposits on the insulator tips, exposed shell surfaces and electrodes, caused by a spark plug with an incorrect heat range, dirty air cleaner, too rich a fuel mixture or excessive idling.
 - Install new spark plugs.



AB0040-A

Fig. 29: Identifying Spark Plug Carbon Fouling
Courtesy of FORD MOTOR CO.

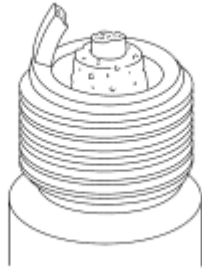
4. Inspect for normal burning.
 - Check for light tan or gray deposits on the firing tip.



AB0039-A

Fig. 30: Identifying Normal Burning Spark Plug
Courtesy of FORD MOTOR CO.

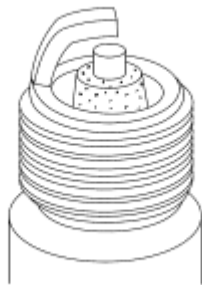
5. Inspect for pre-ignition, identified by melted electrodes and a possibly damaged insulator. Metallic deposits on the insulator indicate engine damage. This may be caused by incorrect ignition timing, wrong type of fuel or the unauthorized installation of a heli-coil insert in place of the spark plug threads.
 - Install a new spark plug.



AB0038-A

Fig. 31: Identifying Spark Plug With Pre-Ignition Condition
Courtesy of FORD MOTOR CO.

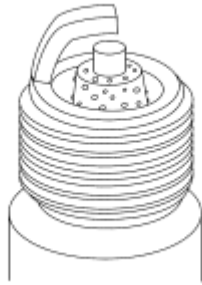
6. Inspect for overheating, identified by white or light gray spots and with a bluish-burnt appearance of electrodes. This is caused by engine overheating, wrong type of fuel, loose spark plugs, spark plugs with an incorrect heat range, low fuel pump pressure or incorrect ignition timing.
 - Install a new spark plug.



AB0042-A

Fig. 32: Identifying Overheated Spark Plug
Courtesy of FORD MOTOR CO.

7. Inspect for fused deposits, identified by melted or spotty deposits resembling bubbles or blisters. These are caused by sudden acceleration.
 - Install new spark plugs.

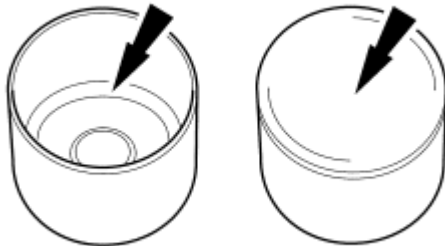


AB0041-A

Fig. 33: Identifying Spark Plug With Fused Deposits
Courtesy of FORD MOTOR CO.

VALVE TAPPET INSPECTION

1. Inspect the valve tappet for damage, especially in the indicated areas. If any damage is evident, inspect the camshaft lobes and valves for damage. Install new components as necessary.



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Fig. 34: Inspecting Valve Tappet
Courtesy of FORD MOTOR CO.

POWERTRAIN/DRIVETRAIN MOUNT NEUTRALIZING

NOTE: Refer to the appropriate service information and procedure for special instructions on loosening and tightening mount fasteners.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING AND LIFTING** article.
2. Loosen, but do not remove, the powertrain/drivetrain mount fasteners.
3. Lower the vehicle.

NOTE: Do not twist or strain the powertrain/drivetrain mounts or damage to the mounts may occur.

4. Start the vehicle and move it in forward 0.6-1.2 m (2-4 ft). Then move the vehicle in reverse the same distance.

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5. Raise and support the vehicle.
6. Tighten the powertrain/drivetrain mount fasteners.
7. Lower the vehicle.
8. Test the system for normal operation.